

Constant power source

X18 (2018) — English translation

The “cunning” source provides constant power P in the external circuit.

A1	Find the current through the source and the voltage across it as a function of time, if it is connected to a resistor with resistance R .	0.50
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A2	Find the current through the source and the voltage across it depending on time, if it is connected to a capacitor of capacity C .	0.50
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Now a circuit is connected to the “cunning” source, the diagram of which is shown in the figure. $R = 5\ \Omega$, $P = 8\ \text{W}$, diode opening voltage $U_0 = 2\ \text{V}$.

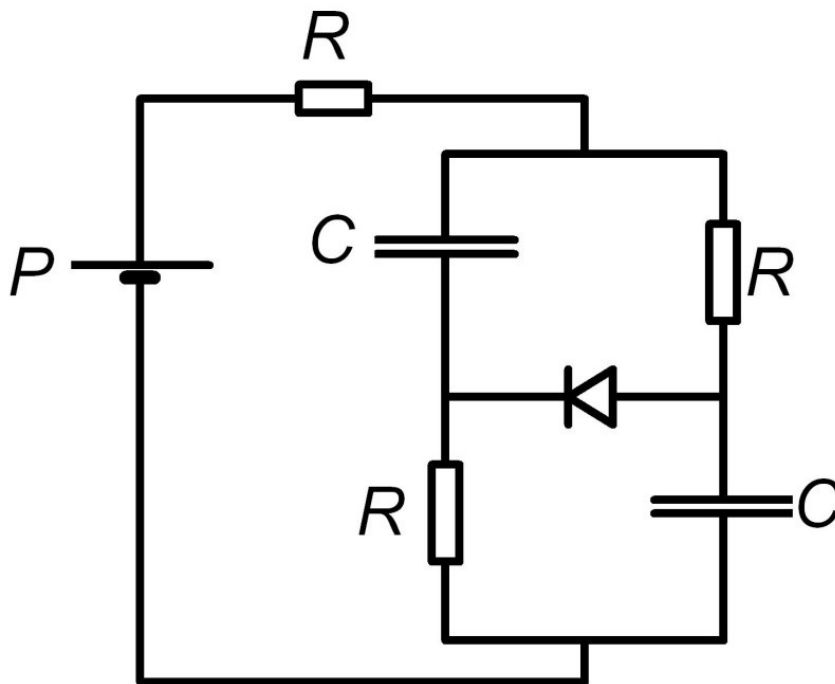


Figure 1: Figure 1.

B1	Find the current and voltage on each element of the circuit at the moment the diode opens.	3.00
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B2	Will steady state be achieved? If so, calculate the current and voltage across each element of the circuit at steady state.	3.00
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Source: <https://pho.rs/p/312>